

Stop Memorizing. Start Understanding Why.

Elaborative Interrogation · Woloshyn, Pressley & Schneider

**"Wiry pulse = Liver/Gallbladder, pain, Phlegm."
Gone by Thursday.**

— Because you memorized the answer. You didn't understand it.

There is a profound difference between **knowing that** something is true and **knowing why** it is true. "Wiry pulse indicates Liver disharmony" is a fact. "Wiry pulse indicates Liver disharmony **because Liver Qi stagnation creates tension — the vessel loses its natural elasticity and feels like a bowstring under your finger**" is knowledge. The first will fade. The second will stick, because it's anchored to a causal mechanism your brain can reconstruct even when direct recall fails.

The Research: Why "Why?" Works

Woloshyn, Pressley, and Schneider (1992) demonstrated that students who asked "why does this make sense?" as they studied significantly outperformed students who used repetition alone — including on delayed tests. **Pressley and colleagues (1992)** extended this finding across science, history, and medical content: when learners generated their own explanations, memory was more durable, transfer to novel problems was stronger, and confusable items were better discriminated.

The mechanism is **elaborative encoding**: each "why" question forces your brain to connect the new fact to existing knowledge, creating multiple retrieval pathways. A fact connected to five other facts is five times harder to forget than a fact that stands alone.

The key finding: Generating your own explanation is significantly more powerful than reading a provided explanation — even if the provided explanation is more accurate. The act of generation, not the correctness of the result, drives the encoding (Pressley et al., 1992).

Applied to the 28 Pulse Qualities

TCM pulse diagnosis is almost perfectly suited to elaborative interrogation because every pulse quality has a beautiful physiological rationale built in. Here's what the difference looks like:

Rote Fact	Elaborated Understanding
Wiry = Liver/GB, pain, Phlegm	

Rote Fact	Elaborated Understanding
	"Liver Qi stagnation = tension in the vessel wall. Like a bowstring — taut, resistant, unable to yield. The same pathological tension that causes flank pain and emotional rigidity shows up in your fingertip as resistance."
Slippery = Phlegm, pregnancy, food retention	"Excess fluid/Phlegm makes the blood viscous and the vessel walls smooth — the pulse feels like it's rolling under your finger, like beads on a string. Pregnancy creates the same slippery quality because Blood volume surges to nourish the fetus."
Hollow = massive Blood loss	"Blood has left the vessel — so the walls are there (surface + deep) but the middle is empty, like pressing a scallion stalk. The body has volume but no substance inside it."
Rapid = Heat	"Heat agitates — it accelerates everything. Heart beats faster to cope with the fever. The same principle: add heat to any system and it moves faster. >90 bpm = Heat is driving the circulation."
Choppy = Blood deficiency, Blood stasis	"Blood is either insufficient (thin, weak flow) or obstructed (rough passage). Either way the pulse is hesitant, uneven, like a knife scraping bamboo — it loses the smooth continuity that well-nourished, free-flowing Blood creates."

The Practice: Three Questions

For any fact in EAM310, run it through three questions before moving on:

1. Why does this make physiological sense?

"Why would Kidney deficiency produce a Deep pulse?" → Kidney is the root — when root Qi is weak, the pulse retreats inward, retreating toward the source of warmth and essence. Deep = the body pulling inward to protect what little it has.

2. How does this connect to what I already know?

"Floating pulse indicates Exterior patterns — how does that connect to the Eight Principles?" → The pathogen is still at the surface (Exterior), not yet penetrated. The body's defensive Qi is fighting there, pushing upward and outward, pulling the pulse up to the surface with it.

3. What would the opposite look like — and why?

"If Floating = Exterior, then Deep = Interior. What does Interior mean for the pulse?" → Pathogen has penetrated, Qi has retreated inward to fight. The contrast clarifies both concepts simultaneously.

"Memory is a web, not a list. Every 'why' you generate is a new strand — and strands don't break the way isolated facts do." — adapted from Pressley et al. (1992)

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